

Empirical Limits of Sparse Attention in Modern Transformers

Abstract

We comprehensively evaluate dense and sparse attention mechanisms on the WikiText-103 benchmark dataset.

Introduction

Recent works advocate sparse attention patterns. We systematically benchmark these approximations against full dense attention.

Method

We evaluate exact dense full-rank attention across identical architectures and learning rate schedules.

Experiments

Experiments are conducted on the WikiText-103 benchmark dataset under rigorous controlled settings.

Results

Sparse attention mechanisms consistently underperform dense attention on the WikiText-103 benchmark, suffering a 4.2 point degradation in perplexity.

Limitations

Dense attention has quadratic time complexity $O(N^2)$ with respect to input sequence length.

Conclusion

Dense attention remains necessary for state of the art representation quality on language modeling tasks.